

PRAQEN — Live Database Migration Report

Complete Database Changes, Tables, Columns & Deployment Architecture

Production Ready

Executive Summary

This document details the complete set of database modifications, 40 new tables, altered core entities, atomic stored procedures, and triggers required to bring the live production server up to date with the latest multi-currency (BTC + USDT TRC-20) architecture.

1. Required PostgreSQL Extensions

Run these commands first in the Supabase SQL Editor:

```
CREATE EXTENSION IF NOT EXISTS "pgcrypto";
CREATE EXTENSION IF NOT EXISTS "vector";
```

2. Inventory of All 40 New Database Tables

#	Table Name	Functional Purpose	Primary Key
1	wallets	Authoritative single source of truth for user balances (BTC + USDT)	id (UUID)
2	user_wallets	HD on-chain deposit addresses & checkpoint tracking	id (UUID)
3	wallet_transactions	Authoritative financial ledger & idempotency locks	id (UUID)
4	deposit_tracking_v2	Blockchain deposit deduplication by tx_hash	id (UUID)
5	deposit_recheck_queue	Operational retry queue for failed explorer checks	(address, currency)
6	realtime_tx_dedupe	WebSocket transaction broadcast deduplication	dedupe_key (TEXT)
7	swap_transactions	Financial ledger for BTC ↔ USDT internal swaps	id (UUID)
8	seller_deposits	Merchant/seller \$200 USDT bond escrow system	id (UUID)
9	reconciliation_flags	Balance anomaly detection flags & audit logs	id (UUID)
10	balance_audit	Immutable audit trail for balance deltas	id (UUID)
11	security_events	Authentication, 2FA, lockout & fraud security logs	id (UUID)
12	security_action_codes	Persistent 2FA short-lived action authorization codes	id (UUID)
13	user_badges	User trust & gamification achievement tracking	id (UUID)
14	user_trust	User trust/block social graph	id (UUID)
15	notifications	User notification center with JSONB payloads	id (UUID)

#	Table Name	Functional Purpose	Primary Key
16	otp_codes	OTP email & phone verification codes	id (UUID)
17	trade_images	Trade chat attachments & payment proofs	id (UUID)
18	support_tickets	Customer support ticketing system	id (UUID)
19	support_messages	Messages within support ticket threads	id (UUID)
20	agent_chat_status	Real-time support agent availability status	user_id (UUID)
21	kb_articles	RAG Knowledge Base with 1024-dim vector embeddings	id (UUID)
22	ai_chat_sessions	AI support bot session management	id (UUID)
23	ai_chat_messages	Conversation history for AI chat assistant	id (UUID)
24	p2p_migration_requests	External reputation import requests (Noones/Binance)	id (UUID)
25	phone_verification_requests	Manual and automated phone verification queue	id (UUID)
26	email_logs	Brevo/SMTP transactional email dispatch logs	id (UUID)
27	admin_audit_log	Administrative action ledger (bans, KYC, roles)	id (UUID)
28	trader_of_week	Active recognition slots for top traders	category (TEXT)
29	trader_of_week_history	Historical archive of all trader awards	id (UUID)
30	suggestions	Community feature request & feedback board	id (UUID)
31	suggestion_votes	Upvotes on community suggestions	id (UUID)
32	dispute_votes	3-of-4 Moderator quorum dispute voting ledger	id (UUID)
33	dispute_comments	Moderator-only discussion threads on disputes	id (UUID)
34	moderator_oaths	Cryptographic/versioned moderator ethics signatures	id (UUID)
35	company_loans	Corporate debt and investor standing loans ledger	id (UUID)
36	company_expenses	Operating expenses tracking	id (UUID)
37	company_staff	Staff directory, payroll & contracts	id (UUID)
38	user_address_migrations	Legacy to new HD treasury address migration ledger	id (UUID)
39	late_deposit_sweeps	Sweeps from deprecated deposit addresses	id (UUID)
40	treasury_migration_transactions	Full audit ledger for on-chain treasury movements	id (UUID)

3. Modified Existing Tables & Columns Added

A. `users` Table (New Columns)

Category	Column Name	Data Type	Default / Notes
Roles & Privileges	<code>is_admin</code>	BOOLEAN	false
	<code>is_moderator</code>	BOOLEAN	false
	<code>is_agent</code>	BOOLEAN	false (Support desk agent)
	<code>is_ceo</code>	BOOLEAN	false (Withdrawal gatekeeper)
KYC & Identity	<code>kyc_status</code>	VARCHAR(20)	'pending' 'approved' 'rejected'
	<code>id_type</code>	VARCHAR(50)	'ghana_card' 'passport' etc.
	<code>id_front_url</code> , <code>id_back_url</code> , <code>selfie_url</code>	TEXT	Document upload URLs
	<code>kyc_submitted_at</code> , <code>kyc_approved_at</code>	TIMESTAMPTZ	Review timestamps
	<code>kyc_rejection_reason</code>	TEXT	Admin review feedback
	<code>kyc_verified</code> , <code>is_id_verified</code> , <code>is_phone_verified</code>	BOOLEAN	false
Security & 2FA	<code>two_factor_enabled</code>	BOOLEAN	false
	<code>two_factor_method</code>	VARCHAR(50)	'email' 'totp'
	<code>totp_secret</code> , <code>reset_password_token</code>	TEXT	Encrypted security tokens
Reputation & Trust	<code>badge</code>	VARCHAR(50)	'BEGINNER' 'PRO' 'EXPERT' 'AMBASSADOR' 'LEGEND'
	<code>has_warning</code> , <code>warning_reason</code> , <code>warned_at</code>	BOOLEAN / TEXT	Trust & Safety moderation warning
	<code>trusted_by_count</code> , <code>blocked_by_count</code>	INT	0
	<code>avg_response_time</code>	NUMERIC	Running average message response time
P2P Migration	<code>p2p_migrated_platform</code>	TEXT	'binance' 'noones' 'paxful'
	<code>p2p_migrated_username</code> , <code>p2p_migrated_feedback</code>	TEXT	External reputation snapshot
	<code>p2p_migration_approved_at</code>	TIMESTAMPTZ	Approval timestamp
Referral & Bonus	<code>referral_code</code> , <code>referred_by</code>	VARCHAR(100) / UUID	Unique referral code & inviter link
	<code>total_referrals</code> , <code>referral_earnings_btc</code>	INT / DECIMAL	Commission accounting

B. `trades` Table (New Columns & Status Update)

- **Added Currency Columns:** `currency` (TEXT, default 'BTC'), `amount_usdt` (DECIMAL(18,6)), `platform_fee_usdt` (DECIMAL(18,6)), `amount_receive_usd` (DECIMAL).
- **Added Trade Tracking:** `trade_ref` (VARCHAR(20) UNIQUE), `payment_proof_url` (TEXT), `payment_proof_at` (TIMESTAMPTZ), `release_tx_hash` (VARCHAR(255)), `buyer_received_btc` (DECIMAL(18,8)).
- **Added Dispute Quorum Columns:** `disputed_by` (UUID), `resolved_via` (TEXT: 'QUORUM' | 'ADMIN_OVERRIDE'), `override_reason` (TEXT), `admin_notes` (TEXT), `dispute_reason` (TEXT), `dispute_resolution` (VARCHAR(50)), `dispute_notes` (TEXT), `disputed_at`, `resolved_by`, `resolved_at`.
- **Constraint Update:** `trades_status_check` updated to include `CANCELLING` intermediate state to prevent cancellation race conditions.

C. `listings` Table (New Columns)

- `asset` (VARCHAR(20) NOT NULL DEFAULT 'BTC' CHECK (asset IN ('BTC', 'USDT'))).
- `gift_card_currencies` (JSONB) — Multi-currency/region configuration array.
- `card_values` (DECIMAL[]), `card_type` (VARCHAR(20)), `face_value` (DECIMAL), `pricing_type` (VARCHAR(20)).

4. Stored Procedures / Atomic PL/pgSQL Functions

1. `praqen_credit_deposit(user_id, currency, amount, onchain_balance, idempotency_key, note)`

Performs atomic deposit crediting: writes transaction ledger entry, increments wallet balance, synchronizes mirrors, advances on-chain checkpoint, and writes audit log inside a single ACID transaction.
2. `praqen_internal_transfer(sender_id, recipient_id, currency, amount, idempotency_key, note)`

Performs deadlock-free atomic user-to-user transfers. Sorts user IDs prior to row-locking, verifies sender balance, updates both wallets, and writes dual ledger entries simultaneously.
3. `praqen_release_escrow` & `praqen_refund_escrow`

Manages atomic escrow release to buyers and dispute refunds to sellers without exposure to concurrency races.

5. Supabase Storage Buckets Required

Bucket Name	Access Level	Purpose
<code>kyc-documents</code>	Private / Authenticated	Government ID, Passport, and selfie uploads
<code>p2p-migration</code>	Public	Reputation screenshot proofs from external platforms
<code>trade-proofs</code>	Authenticated	Fiat payment receipts and trade confirmation images
<code>avatars</code>	Public	User profile avatars

6. Deployment Execution Order for Main Developer

1. Execute master script: `RUN_COMPLETE_DATABASE_SETUP.sql`
2. Execute dispute quorum schema: `dispute_voting_tables.sql`
3. Execute ERP books & staff: `company_books_tables.sql` and `company_staff_table.sql`
4. Execute team features: `team_features_tables.sql`
5. Execute RAG Vector schema: `RUN_THIS_IN_SUPABASE.sql`
6. Initialize user wallets & assign admin role.